BIG MODULAR MULTI–MIX – carrier (jack board), S3. Digital 4–channel mixer: 4 IN + 2 RETURN, SEND A / SEND B / MAIN OUT, 15 knobs read by the MCU (MIDI CC), USB audio + MIDI, PCM3168A at 48 kHz, clock link. Circuit intent: docs/research.md

POWER (as modules/usb–audio–max): +12V –> D2 –> +12V_F (output op–amps; R1||R2 16.5R –> Q1 BCP56 (U1A LM358 loop, ref U9 REF5050) –> +5VA; FB2 + R3 –> U2 TPS562208 FCCM buck –> L1 –> +3V3 –> FB1 –> +3V3_C –> R21 –> +3V3_OSC); –12V –> D1 –> –12V_F. +12 V ~0.32 A max, –12 V ~10 mA.

DC REFERENCE (kept from modules/usb–audio–max): +5VA sets the codec gain (DAC 1.6 x VCC, ADC 0.2 x VCC):
U9 REF5050A –> R6 10k / C19 1u –> V5_SET –> U1A LM358 –> R9 –> Q1 BCP56 –> R8 –> +5VA;
R7 10k senses +5VA at the codec; C37 22n + R16 47k = PI compensation. Q2 limits at ~0.3–0.35 A.

PCM3168A at 48 kHz: I2C 0x44 (MODE = DGND, ADR0 = GND, ADR1 10k to GND). SCKI 24.576 MHz = 512 fS, single–rate 24–bit
I2S–mode TDM, BCK 256 fS = 12.288 MHz: DOUT1 = ADC1–6, DIN1 = DAC1–8 (DOUT2 open, DIN2–4 GND).
DAC slots: VOUT5 SEND A, VOUT3 SEND B, VOUT4 MAIN, VOUT1 USB 1, VOUT2 USB 2 (net DACn = output n).
Stand–alone / chain master: ADC = clock master, DAC slave on the tied BCK/LRCK. Clock slave (LINK_N low):
ADC + DAC slaves, U11 feeds the link SCKI, U13 the link BCK/LRCK. Firmware: SEAD = 1, REVAD = 1.

MCU BOARD LINK: J12/J13 1x12 2.54 mm male strips (plastic on the carrier F side);
the MCU board (multi–mix–mcu, J112/J113) solders on top. BOOT/RESET and SWD on the carrier (B side):
hold BOOT, press RESET = CH32V307 ROM bootloader on the USB–C. I2C pull–ups are on the MCU board.

+3V3 BUCK, kept out of the audio:
– only the digital rail is switched; +5VA stays linear
– U2 is the FCCM TPS562208: fixed ~580 kHz at every load
  (TPS562201 pulse–skips into the audio band: do not substitute)
– FB2 + R3 2.2R + C6/C7 keep its switching current off +12V_F
– L1 molded; FB1 filters the codec VDD / clock logic, R21/C43 Y1

OUTPUTS (x5): DAC1 SEND A (J1, VOUT5), DAC2 SEND B (J2, VOUT3), DAC3 MAIN OUT (J5, VOUT4) on U7A–C; DAC4 USB 1
(J9, VOUT1), DAC5 USB 2 (J10, VOUT2) on U8 (host–assigned outs, bottom right); VOUT6–8 open. Difference amplifier on +/–12 V (15k / 27k || 47p,
gain 1.8) –> 2k || 2k (1k / 0.5 W) –> jack. Full scale +/–7.2 V.

PWR_FLAG: GND/+12V/–12V at J8; +12V_F/–12V_F behind D2/D1; V5_IN behind R1||R2; V3_IN behind FB2 + R3; +3V3 behind L1; +5VA at R8; +3V3_C behind FB1; +3V3_OSC behind R21; +3V3_POT behind R62.

INPUTS (x6: CH1–4, RTN A/B): jack (TN = NORM) –> 10u –> 100k –> inverting driver on +5VA (–0.2, + at VCOM_REF)
–> 470R / 2.2n C0G –> VINn+ (single–ended). +/–7.07 V = 1 Vrms ADC full scale. 100k, AC–coupled (modules/4mix D3).

CLOCK LINK (keyed 2x5 IDC daisy chain, straight ribbon CLK OUT –> next unit CLK IN; master = the unit with CLK IN open):
1 DET | 2 NC | 3 GND | 4 SCKI | 5 GND | 6 BCK | 7 LRCK | 8 GND | 9 NC | 10 NC (Eurorack power cable safe)
Clock only (sample–locked USB multitrack): each unit keeps its own USB and outputs. Up to 4 units until bench–tested.
LINK_N low (CLK IN connected) = clock slave: U11 takes SCKI and U13 BCK/LRCK from the link, by hardware;
U14A/U12 repeat them out of CLK OUT to the next unit.

NORM: all six jack normals on one node; NORM_PROBE (PA8) pseudo–random bits via 100k/1k = 33 mV: the firmware
correlates it in each ADC channel to tell patched from unpatched (RTN A/B unpatched –> returns from USB).

VCOM (+5VA): U6C buffers VCOMAD –> 100R / 10u –> VCOM_REF (VIN1–6 – and the driver + inputs). U6D, U7D spare.

KNOBS: 15 x RV09 50k B (modules/4mix references) as DC dividers on +3V3_POT; nothing on them carries audio.
Wiper 100n –> U10 74HC4067 (I0–I14, I15 = GND) –> R61 1k –> POT_ADC (MCU PC1 = ADC_IN11); S0–S3 = PC2, PC3, PA0, PA1.
The firmware scans them, smooths with hysteresis and sends MIDI CC (VOL as CC7 per channel).

Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S4 – digital 4–channel mixer: PCM3168A at 48 kHz, 4 IN + 2 RTN, SEND A/B + MAIN +
Big Modular
Sheet: /
File: multi–mix.kicad_sch
Title: Big Modular – MULTI–MIX carrier (jack board)
Size: A1
Date:
Rev: S4
KiCad E.D.A. 10.0.6
Id: 1/1